

ARGHAVAN (NOVA) EBRAHIMI

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SUMMARY

PhD researcher in Human-Centered Computing specializing in mixed-methods research spanning qualitative, quantitative, and computational approaches. Designs and evaluates studies from stakeholder interviews and focus groups to survey design and A/B testing to large-scale, machine-learning-assisted analysis of interaction data, translating empirical findings into design frameworks and actionable recommendations. Builds Python-based data pipelines and applies vision-language models to large, multi-platform research datasets on high-performance computing infrastructure. Strong communicator and cross-disciplinary collaborator, experienced in presenting complex findings to both technical and non-technical stakeholders.

CORE SKILLS & TOOLS

Research & Analysis: Field and lab study design, usability testing, semi-structured & structured interviews, focus groups, ethnography, tree testing, concept/prototype testing, personas, journey maps, service blueprints, thematic & content analysis, survey design, A/B testing, statistical analysis (descriptive & inferential)

Core Strengths: Human-centered design thinking, human-machine interaction, insight synthesis & data storytelling, user-interaction & flow mapping, stakeholder engagement & cross-functional collaboration, communication & presentation, active listening, empathy-driven problem-solving & critical thinking

Design & Prototyping Tools: Figma, FigJam, Balsamiq, Adobe Photoshop, Adobe Illustrator, Rhinoceros 3D, Miro, Airtable, Qualtrics

Prototyping: Rapid, low- and high-fidelity prototyping, wireframing, paper prototyping

Programming & AI/ML: Python; data pipeline development on high-performance computing (HPC) clusters; vision-language models (Qwen, Gemini) for large-scale image and data analysis

Data Visualization: Tableau; Python (Matplotlib, Plotly)

PROFESSIONAL EXPERIENCE

Teaching Assistant | Human-Centered Computing Lab, UNC Charlotte

Charlotte, NC | Aug 2021 – Present

- Led mixed-methods user research, including 14 educator interviews and 9 student focus groups across platforms such as Second Life, Gather.Town, Virbela, and Minecraft Education, to identify design patterns for virtual learning environments, synthesized into the Socio-Spatial Embodiment (SSE) design framework and published across FIE, ACM Interactive Surfaces & Spaces, and Virtual Worlds Journal.
- Designed a psychometric instrument (the SSE Scale) measuring students' integrated sense of place, presence, and community in virtual environments.
- Built and maintained a Python-based computational analysis pipeline combining computer vision, vision-language models (Qwen, Gemini), and machine learning clustering on a high-performance computing (HPC) cluster to analyze approximately 7,000 virtual-world interface images across seven platforms, producing a systematic, data-driven design-pattern taxonomy.
- Applied interdisciplinary expertise spanning UX research, HCI, human factors, architectural design, and educational technology to translate empirical research findings into actionable design recommendations for diverse stakeholders, including educators and platform designers.
- Led teaching-assistant teams across Human-Computer Interaction, Rapid Prototyping, and Interaction Design Studio courses, instructing students in UX deliverables such as personas, journey maps, and service blueprints while managing logistics and conducting independent doctoral research.

Research Assistant | Integrated Design Research Lab, UNC Charlotte

Charlotte, NC | Jan 2021 – May 2021

- Researched spatial typologies bridging human and machine perception, prototyping and programming a scanner robot using Raspberry Pi, proximity sensors, and LiDAR.

Research Assistant | Studio Dickey Lab & Urban Synergetic Lab, UNC Charlotte

Charlotte, NC | Aug 2019 – Jan 2021

- Designed and prototyped origami acoustic structures using Rhinoceros 3D, Grasshopper, and Pachyderm Acoustic; presented and animated results using Grasshopper, Photoshop, and InDesign.
- Researched mechanical computer shape grammar using a Logic Gate Simulator.

Research Assistant | High-Performance Architecture Lab, Tarbiat Modares University

Tehran, Iran | Sep 2016 – Jul 2019

- Researched and published papers on architectural shape grammar methodologies.

Interior Architect | Pasar Interior & Decoration Design Office

Tehran, Iran | Oct 2014 – Jun 2016

- Designed residential and commercial interiors and oversaw implementation from concept through completion.
- Produced client presentations, renderings, and cost estimates using Rhinoceros 3D, AutoCAD, 3D Max, and Photoshop.

Early Design Experience

Tehran, Iran | 2010 – 2014

- Interior Designer & Architect, Shabahang Interior Design Office (May – Oct 2014): co-designed commercial project plans and oversaw implementation.
- Junior Architect, Emarat-e-Khorshid (EK) Consulting Engineers (Aug 2013 – May 2014): designed a commercial complex within a historical context in Tabriz using AutoCAD and 3D Max.
- Architectural Design Intern, ERA Consulting Engineers (Apr – Sep 2010): drafted project plans and produced renderings using AutoCAD and Rhinoceros 3D.

EDUCATION

Ph.D. in Computing and Information Systems | University of North Carolina at Charlotte

Human-Centered Computing Lab | Aug 2021 – Present (dissertation defense expected Fall 2026)

- Research: design of virtual world interfaces and learning environments.

Dual M.S. in Information Technology & Design Computation | University of North Carolina at Charlotte

May 2021

- Relevant coursework: Human-Centered Design, Web Application Design, AR User Experience Design, IT Project Management, Computational Design Theory, AR Design Studio, Digital Fabrication.
- Thesis: "Revealing the Hidden Layers in an Environment: A Case Study of the Visualization of Coronavirus Contamination and Navigation in a Hospital" (an Augmented Reality project).

M.S. in Architectural Engineering | Rouzbahan Institute of Higher Education

Sari, Iran | Feb 2013

- Coursework: Design Theory, Urban Design, Parametric Design, Design Patterns.

B.S. in Architectural Engineering | Tabriz Art University

Tabriz, Iran | Sep 2010

- Coursework: Architectural Design, Prototyping, Design Theory, Urban Design, 3D Modeling, Product Design.

CERTIFICATIONS

- Foundations of User Experience (UX) Design | Google
- Start the UX Design Process: Empathize, Define, and Ideate | Google

SELECTED PUBLICATIONS

8 peer-reviewed publications (2022–2025) on spatial design, embodiment, and virtual environments:

- Ebrahimi, A., Maher, M.L. & Ramaprasad, H. (2025). "Successful Teaching Strategies in Virtual World Education for Engineering and Computing." In Proceedings of Frontiers in Education 2025, Nashville, TN, USA.
- Ebrahimi, A., & Ramaprasad, H. (2025). "Successful Practices for Avatars in Virtual Learning: Perspectives from Experienced Educators." In Proceedings of IEEE Digital Education and MOOCS Conference, Boca Raton, FL, USA.
- Ebrahimi, A., & Ramaprasad, H. (2025). "Student Experiences in Online Learning Environments: A Comparative Study of Virtual Worlds and Video Conferencing Platforms." In Proceedings of IEEE Digital Education and MOOCS Conference, Boca Raton, FL, USA.
- Ebrahimi, A., & Ramaprasad, H. (2025). "Advantages of Virtual Worlds Versus Video Conferencing Tools for Online Education: A Comparative Study." In Proceedings of IEEE Digital Education and MOOCS Conference, Boca Raton, FL, USA.
- Ebrahimi, A., & Ramaprasad, H. (2025). "Interaction Design Strategies for Socio-Spatial Embodiment in Virtual World Learning." Virtual Worlds (Vol. 4, No. 3, p. 30). MDPI.
- Ebrahimi, A., & Maher, M.L. (2024). "Designing Presence and Place: A Framework for Engaging Student Interaction in Desktop Virtual World Learning Environments." In Proceedings of Frontiers in Education 2024, Washington, DC, USA.
- Ebrahimi, A. (2023). "Empowering Online Learning: AI-Embedded Design Patterns for Enhanced Student and Educator Experiences in Virtual Worlds." In Proceedings of ACM Interactive Surfaces and Spaces 2023, Pittsburgh, PA, USA.

- Najjar, N., Ebrahimi, A., & Maher, M.L. (2022). "A Study of the Student Experience in Video Conferences and Virtual Worlds as a Basis for Designing the Online Learning Experience." In Proceedings of Frontiers in Education 2022, Uppsala, Sweden.